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Automated quantification of abdominal aortic calcification using 3D nnU-Net: a novel approach to assess AAA rupture risk

Yuan-Lin Luo^{1,2†}, Yi-Fan Liu^{1†}, Zhi Huang¹, Chu Wang¹, Ling-Yue Zhang³ and Shui-Chuan Huang^{1*}

Abstract

Background Abdominal aortic aneurysms (AAA) pose a serious rupture risk, heightened by aortic calcification. Traditional calcification scoring methods are slow and require expertise. This study aims to construct a convolutional neural network (nnU-Net) model for automatic quantification and segmentation of abdominal aortic calcification from a single CTA scan.

Methods This retrospective study included 100 patients who underwent abdominal aortic CTA between January 2018 and October 2023, meeting specific inclusion criteria. Vessel and calcification segmentation were manually scored by two physicians, and an nnU-Net deep learning model was developed to automate calcification measurement. Model performance was assessed using Dice scores. Agreement between manual and model-based scoring was assessed using Spearman rank correlation and Bland-Altman analysis.

Results The nnU-Net model achieved median Dice scores of 93.60% for blood vessels and 81.06% for calcification. Average Dice scores were $92.37 \pm 4.87\%$ for blood vessel segmentation and $81.03 \pm 5.11\%$ for calcified plaque. The model's Agatston scores correlated closely with manual scores (Spearman's $\rho = 0.969$), with a mean difference of -229.51 (95% limits of agreement: -6003.92 to 5544.90). The model's evaluation time was also shorter than manual scoring (112 ± 4.4 s vs. 3796 ± 6.6 s, $p < 0.001$).

Conclusion The nnU-Net-based model shows potential as an automated tool for accurately segmenting and quantifying abdominal aortic calcification, offering comparable results to manual scoring with significantly reduced evaluation time. This approach may assist in more efficient assessment of AAA rupture risk, supporting clinical decision-making in patient management.

Keywords Abdominal aortic aneurysm, Convolutional neural network (nnU-Net), Calcification, Rupture

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Introduction

In developed countries, the prevalence of abdominal aortic aneurysm (AAA) ranges from 4–8% [1], increasing with age and primarily affecting men. Although often asymptomatic, AAA rupture can result in severe outcomes, with an overall mortality rate exceeding 80% [2]. The primary predictor of rupture risk is the maximum diameter (DMax) of the aneurysm [3, 4]. Clinically, an AAA diameter of ≥ 5.5 cm is the standard threshold for recommending endovascular aneurysm repair (EVAR) [5, 6]. However, studies indicate that 2–3% of ruptured AAAs measure less than 5 cm, highlighting the need for additional factors to improve rupture risk assessment [7, 8]. Recent research underscores the relevance of parameters such as wall stress and aortic calcification levels, with higher calcification scores significantly linked to symptomatic AAAs and rupture events [9–16].

Current methods for assessing aortic calcification, including the Agatston scoring system and the Callister method, require manual identification and annotation of calcific lesions in individual image slices [11]. These processes are time-consuming and demand a moderate level of expertise [17, 18]. While semi-automated and fully automated techniques have been introduced for coronary calcium segmentation, similar methods for quantifying aortic calcification using computed tomography angiography (CTA) are still relatively uncommon [19–21].

As a self-configuring deep learning framework, nnU-Net has demonstrated effectiveness and versatility in medical image segmentation. By reducing the need for task-specific adjustments, nnU-Net has been successfully applied in the diagnosis and segmentation of cardiovascular diseases [22]. For instance, a U-Net-based deep learning model achieved an average Dice similarity coefficient (DSC) of 0.889 when segmenting coronary arteries from invasive angiography, showcasing its clinical potential [22]. However, its application in automating the assessment of abdominal aortic calcification in CTA images remains unexplored [18, 23, 24].

This study developed a nnU-Net-based model for fully automated segmentation and quantification of abdominal aortic calcification from single CTA scans and evaluated its performance.

Methods

Materials and methods

This retrospective, observational study was conducted at the Division of Vascular and Interventional Radiology, Department of General Surgery, Nanfang Hospital, Southern Medical University. The study analyzed data from 100 patients diagnosed with abdominal aortic aneurysm (AAA) who underwent computed tomography angiography (CTA) between January 2018 and October 2023.

Patients were included if they had a diagnosis of AAA confirmed by clinical and imaging criteria, CTA scans acquired according to standardized imaging protocols, and complete clinical baseline and imaging data available. Patients were excluded if they had previously received endovascular or open surgical repair, had severe imaging artifacts or missing slices on CTA scans, had no calcified plaques in the abdominal aorta, had incomplete clinical data, severe artifacts: motion/beam hardening affecting $> 30\%$ slices, and incomplete slices: > 3 mm z-axis gap or corrupted DICOM data. A total of 187 patients were screened; 87 excluded (37 prior repairs, 28 severe artifacts, 15 no calcific plaques, 7 incomplete data).

Ethical approval and informed consent

The study protocol was reviewed and approved by the Southern Medical University Health Research Ethics Committee (Approval ID: NFEC-202504-K3-01). All participants provided written informed consent prior to data collection.

Clinical and imaging data collection

Clinical baseline data included age, gender, body mass index (BMI), smoking history, comorbidities (hypertension, diabetes, hyperlipidemia, chronic kidney disease, cardiovascular disease), maximum lumen diameter of the AAA, and sociodemographic information (marital status). Imaging data were obtained from CTA scans performed using standard protocols.

Manual annotation and calcification scoring

CTA images were segmented and annotated by two experienced vascular surgeons using 3D Slicer® software (version 5.2.2). Calcific lesions were defined as regions with attenuation values ≥ 130 Hounsfield units (HU) and a minimum area of 1 mm^2 . Manual annotation involved frame-by-frame verification of calcification and arterial lumen markings within the anatomical region extending from the celiac trunk to the iliac bifurcation. Calcification scoring was performed using the Agatston method, which weights lesion area by density: 130–199 HU scored 1, 200–299 HU scored 2, 300–399 HU scored 3, and ≥ 400 HU scored 4.

Development of the nnU-Net model

The nnU-Net deep learning framework was employed for automated segmentation and quantification. The model automatically configured parameters based on dataset characteristics, including voxel spacing, image size, and modality. Five-fold cross-validation was used to train and validate the model, with post-processing techniques such as connected component analysis applied to improve segmentation accuracy. The segmentation pipeline included preprocessing, model training, and post-processing

steps, and outputted segmentation masks and Agatston scores for calcific lesions. Preprocessing: “Resampling to isotropic 1 mm³ voxels, intensity clipping (0–95% percentile), z-score normalization.” Post-processing: “Largest connected component for vessels; calcification masks filtered by HU ≥ 130 and area ≥ 3 voxels; morphological closing (sphere kernel, r = 2 mm).” Model development utilized PyTorch 2.2 with CUDA 12.4 acceleration and nnU-Net v 2.1.1 - the current gold standard for medical image segmentation. We implemented MONAI 1.3.1 for specialized 3D medical imaging preprocessing and TorchIO 0.3.10 for advanced data augmentation. Batch-Generators (v0.25) facilitated optimized patch-based training for large volumetric data, while SimpleITK (v2.3.0) enabled DICOM handling and spatial transformations including anisotropic resampling and coordinate system conversions.

Performance evaluation

Segmentation performance was evaluated using the Dice similarity coefficient for blood vessels and calcified plaques. Agreement between manual and model-based scoring was assessed using Spearman rank correlation and Bland-Altman analysis. The Dice score, used to assess segmentation accuracy, quantifies the overlap between predicted and actual masks. It considers true positives (correctly identified voxels), false positives (incorrectly identified positives), and false negatives (incorrectly identified negatives). Higher Dice scores denote better alignment and precision, with scores of 80% or more indicating substantial similarity between manual annotations and model-generated segmentations. Hausdorff distance (HD) was computed to quantify the maximum boundary error between predicted and manual segmentations. We report the 95th percentile HD (HD95) to reduce

sensitivity to outliers. Volume similarity (VS) was calculated as $1 - |V_{pred} - V_{gt}| / V_{gt}$, where V_{gt} is the ground truth volume. Risk stratification was evaluated through a 3 × 3 confusion matrix (Low / Moderate / High). Agreement was quantified using Cohen’s kappa (κ), with κ > 0.6 considered substantial [4]. Precision, recall, and F1-score were calculated per category. Augmentation: Rotation (± 15°), scaling (± 10%), Gaussian noise (σ = 0.05). Training: 100 epochs per fold, early stopping if validation loss plateaued > 10 epochs.

Statistical analysis

Statistical analyses were performed using SPSS version 26.0 (IBM Corp, Armonk, NY, USA). Continuous variables were expressed as mean ± standard deviation (SD). Bland-Altman plots were used to assess agreement between manual and automated scoring methods, with 95% confidence intervals reported. PyTorch2.2 was used to develop the model.

Results

Clinical characteristics

The whole process of this study was presented (Table 1). The study cohort comprised 100 patients with abdominal aortic aneurysms (93% male; mean age 72.5 ± 7.8 years), demonstrating typical AAA demographic patterns. Key cardiovascular risk factors included hypertension (82%), current/former smoking (62%), and renal insufficiency (54%), while diabetes (12%) and hyperlipidemia (16%) were less prevalent. The mean maximum AAA diameter was 5.1 ± 1.2 cm, with severe calcification burden evidenced by Agatston scores (20,431 ± 8,972). Participants exhibited normal BMI (23.5 ± 3.5 kg/m²) and universal marital status (100% married), with predominantly rural residence (59% vs. 37% urban; 4% self-funded). These baseline characteristics align with contemporary Chinese AAA registry data.

Table 1 Clinical characteristics

Characteristic	Value (n = 100)
Age (years)	72.5 ± 7.8
Gender, No. (%)	100
Male/	93 (93%)
Female	7 (7%)
Max AAA Diameter (cm)	5.1 ± 1.2
Hypertension	82 (82%)
Diabetes	12 (12%)
Current/Former smoker	62 (62%)
Agatston Score	20,431 ± 8,972
BMI (kg/m2)	23.5 ± 3.5
Hyperlipidemia, No. (%)	16 (16%)
Renal insufficiency	54 (54%)
Location, No. (%)	100
Urban	37 (37%)
Rural	59 (59%)
Self-funded	4 (4%)
Marriage status, No. (%)	100 (100%)

Comparison between manual and model-based segmentation

Figure 1 (A-C) outlines the manual segmentation procedure for blood vessels and calcified regions, whereas Fig. 1 (D-E) presents the automatic segmentation conducted by the nnU-Net model on the same slice. Despite the intricate nature of the branching vessels, soft tissues, and organs near the abdominal aorta, the nnU-Net model successfully localized the target areas without misclassifying neighboring structures. The segmentation produced by the model closely aligned with the manual segmentation, avoiding both over-segmentation and under-segmentation. Furthermore, the 3D reconstruction of the segmented images (Fig. 2A-B) confirmed that the model maintained structural integrity and retained critical information, meeting the expected standards.

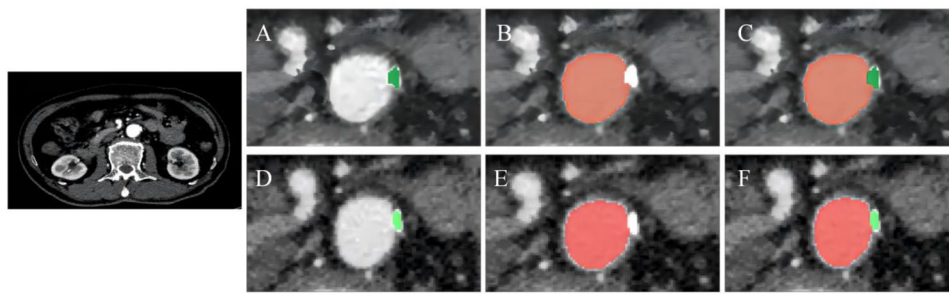


Fig. 1 Comparison of manual and model segmentation: **A-C.** Manual segmentation, **D-F.** Model segmentation

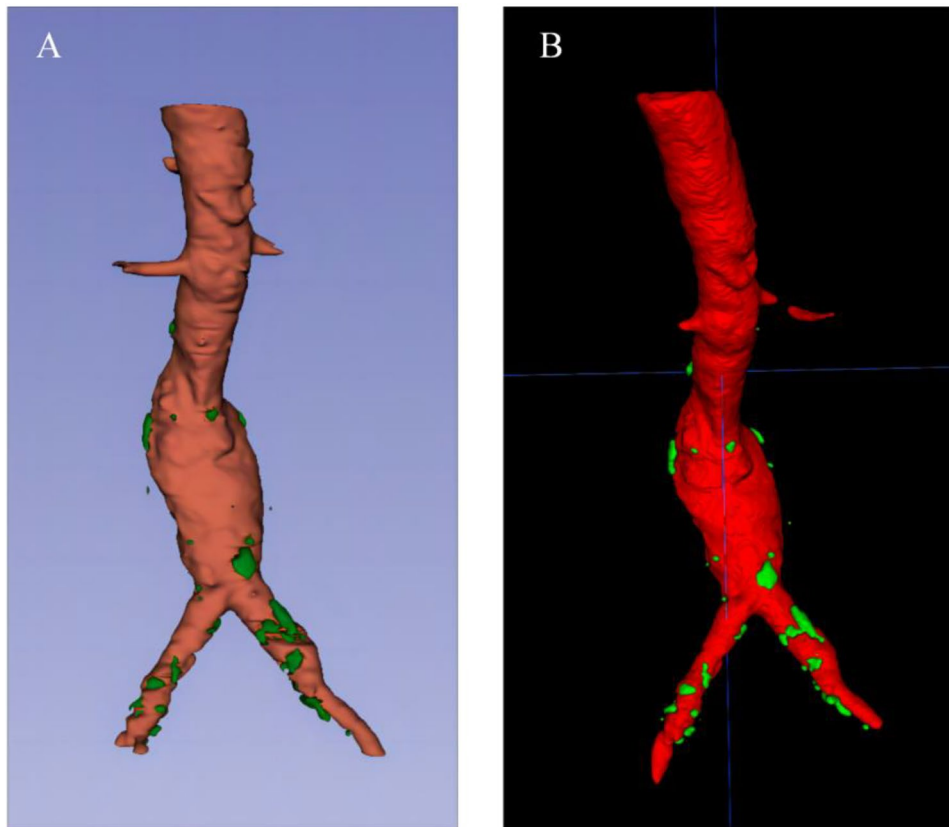


Fig. 2 3D reconstruction of artificial and model segmentation: **A.** Manual reconstruction, **B.** Model reconstruction

Model convergence curve

Within convolutional neural networks, one epoch signifies a full cycle of forward and backward propagation through the entire dataset, enabling the model to learn from all training samples. Through successive epochs, the model gradually adjusts to the training data, moving from underfitting to an optimal fit, while its ability to generalize is tested on the validation set. Selecting the appropriate number of epochs is essential; an excess can cause overfitting, while too few can lead to underfitting. In this study, 5-fold cross-validation was utilized, allowing nnU-Net to automatically extract ROI data features during each fold, thereby continuously improving model performance. The loss function curves from the five

cross-validation runs, as shown in Fig. 3, reveal that by Epoch 108, the model's loss function stabilized in both the training and validation sets, with values smoothly converging toward 1. The final cross-validation curve (Fig. 3E) indicates that both training and validation losses converged with negligible differences, suggesting a well-fitted and stable model.

Model segmentation performance

According to Fig. 4, Dice scores for blood vessel segmentation ranged from 65.17 to 97.77%, with a median of about 93.60%. For calcification segmentation, scores varied from 64.82 to 89.81%, with a median of approximately 81.06%. Table 2 reveals that the model achieved an

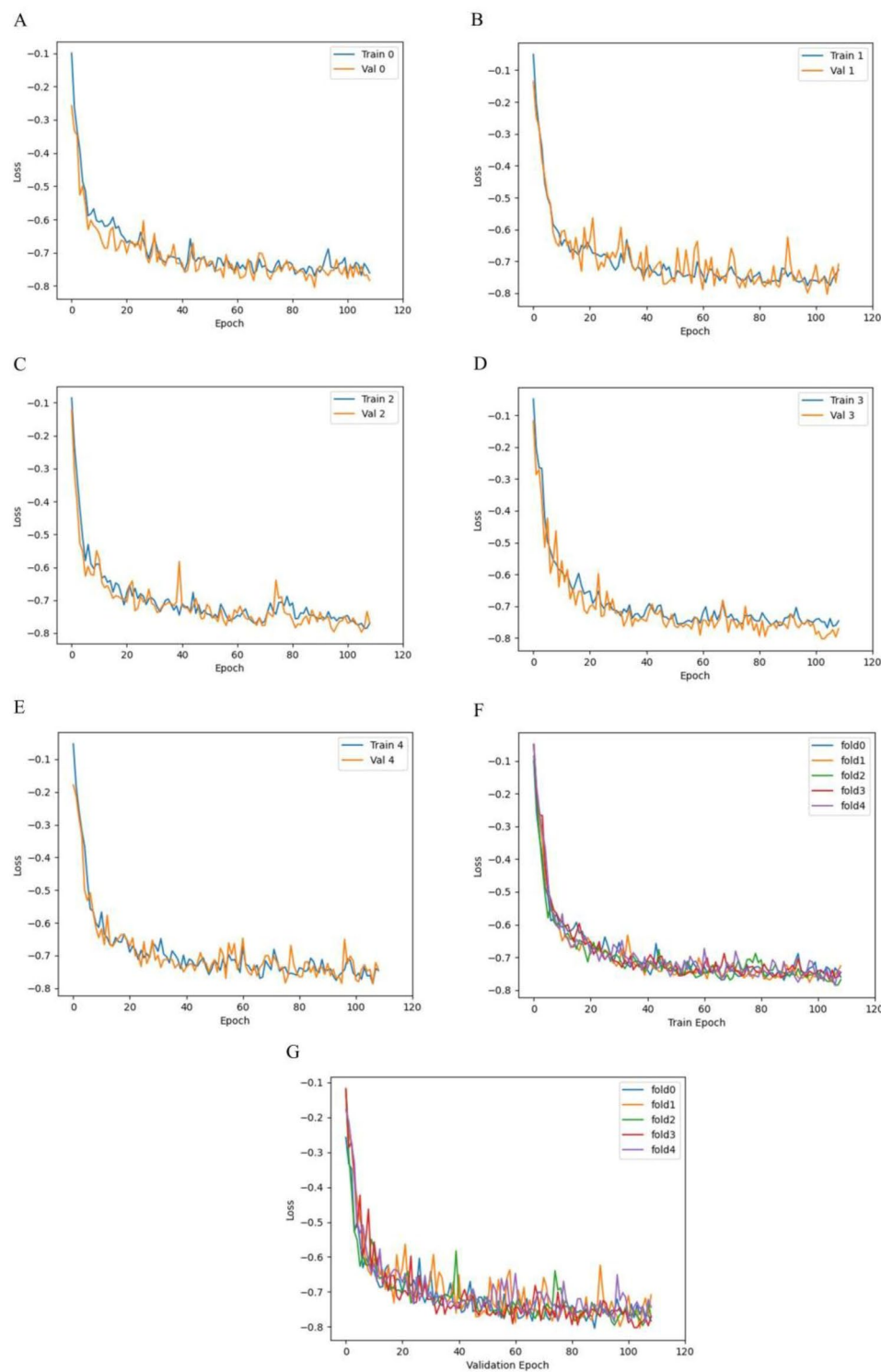


Fig. 3 **A.** First cross-validation loss function curve; **B.** Second cross-validation loss function curve; **C.** Third cross-validation loss function curve; **D.** Fourth cross-validation loss function curve; **E.** The fifth cross-validation loss function; **F.** 5-fold loss function curve in training set; **G.** 5-fold loss function curve in validation set

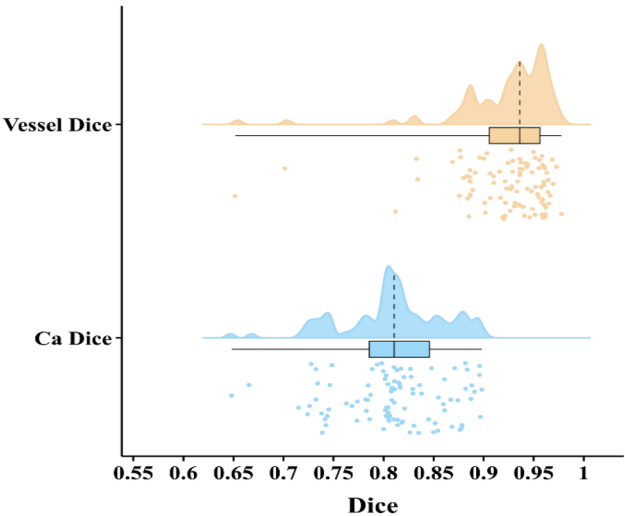


Fig. 4 Dice distribution of blood vessels and calcification

Table 2 Dice score, HD and VS		
Sample (N=100)	Blood Vessels	Calcified Plaques
Dice score (%)	92.37±4.87	81.03±5.11
HD	2.4 mm [IQR:1.8-3.1]	3.1 mm [IQR:2.5-4.0]
VS	94.2%±3.1%	86.7%±4.8%

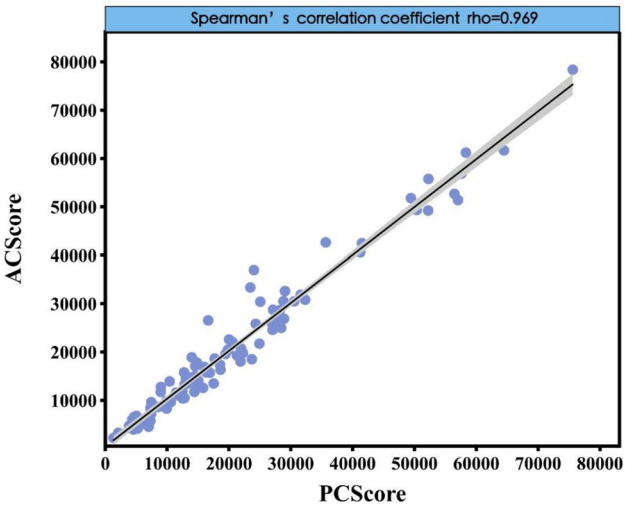


Fig. 5 Correlation between manual and model score (PCScore: Physical calcification score; ACSore: Automated calcification score)

average Dice score of $81.03 \pm 5.11\%$ for calcified plaques and $92.37 \pm 4.87\%$ for blood vessel segmentation, demonstrating robust performance across both categories. The median HD95 was 2.4 mm (IQR: 1.8–3.1 mm) for vessels and 3.1 mm (IQR: 2.5–4.0 mm) for calcification. Higher HD values in calcification segmentation occurred at plaque edges with low contrast (Fig. 5C, arrow). Volume agreement was high for vessels (VS: $94.2\% \pm 3.1\%$) and calcification (VS: $86.7\% \pm 4.8\%$). Lower VS in calcification

reflected occasional under-segmentation of small plaques ($<5\text{ mm}^3$).

Differences between manual and model-based scoring

To evaluate the consistency between manual segmentation scores and nnU-Net predicted Agatston scores, Spearman’s rank correlation coefficient was calculated, resulting in $\rho = 0.969$, which denotes a strong correlation between the two methods (Fig. 5). Bland-Altman analysis (Fig. 6) indicated minimal discrepancies between manual and model-predicted Agatston scores, with a mean difference of -229.51 (95% limits of agreement: -6003.92 to 5544.90). The box plot (Fig. 7) illustrated that the medians of the manual and model scores were virtually identical, with minimal variance. The model achieved 79% overall accuracy ($\kappa = 0.71$) in risk classification (Fig. 8). Performance varied by category: High Risk showed highest F1-score (83.3%), while Moderate Risk had lower recall (73.0%). Most misclassifications occurred between adjacent categories (e.g., 7 Low→Moderate, 6 Moderate→High), reflecting threshold ambiguity. All statistical comparisons inherently accounted for paired measurements, as each subject served as their own control for manual versus automated scoring methods.

Discussion

This study presents a novel fully automated model based on nnU-Net deep learning, designed to identify and quantify abdominal aortic calcification in patients using computed tomography angiography (CTA). The results indicate that the calcification areas identified and quantified by the nnU-Net model correlate highly and consistently with manual segmentation scores. Additionally, the model efficiently detects calcified regions within the abdominal aorta, achieving good segmentation accuracy and automatically calculating the total calcification score. This significantly reduces analysis time compared to manual annotation, paving the way for the clinical application of AI-based AAA calcification scoring technology.

Our focus was on quantifying calcification levels within the abdominal aorta. The mean calcification scores from manual segmentation and the AI model were 20431.30 and 20660.81, respectively, both suggesting that the average abdominal aortic calcification score for AAA patients exceeds 20,000—substantially higher than that of a normal abdominal aorta. This elevated level of calcification in AAA patients hints at a potential link between abdominal aortic aneurysms (AAA) and calcification [15, 25]. AAA is a life-threatening condition primarily treated with endovascular therapy or open surgical repair [5, 6]. Guidelines from the Society for Vascular Surgery and the European Society for Vascular Surgery recommend treatment decisions based on balancing surgical risk with the risks of AAA growth and rupture [2, 25].

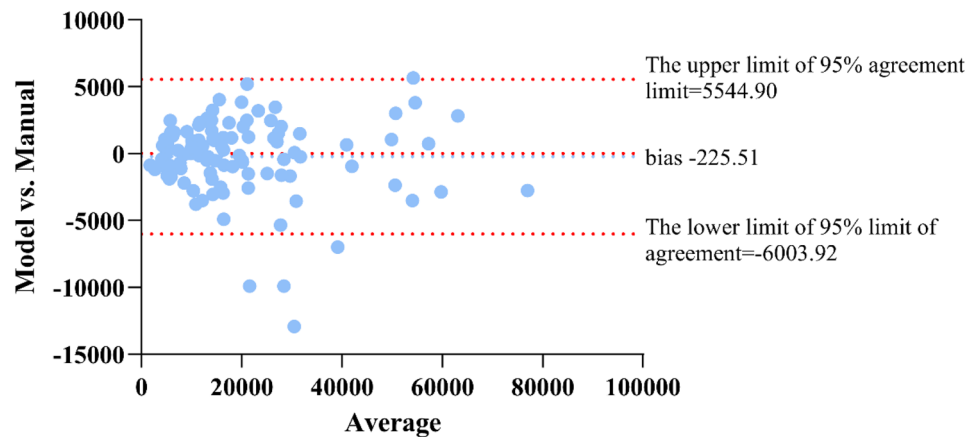


Fig. 6 Bland-Altman plots of manual and model scores

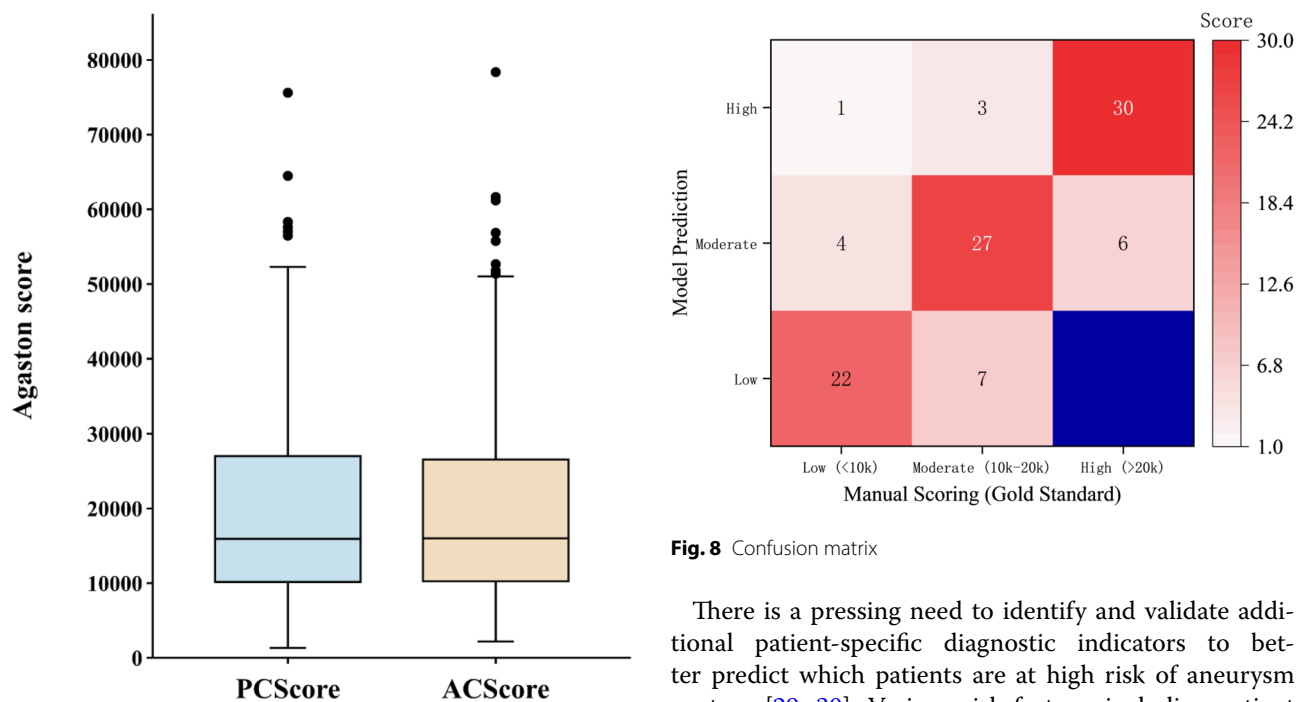


Fig. 8 Confusion matrix

Fig. 7 Distribution of manual and model agatston scores (PCScore: Physical calcification score; ACSore: automated calcification score)

Currently, maximum AAA diameter is recognized as an independent predictor of rupture risk, where small aneurysms (diameter <5.0 cm for women or <5.5 cm for men) and slow-expanding aneurysms (<0.3 cm/year) generally indicate a lower risk [25–28]. In our cohort (mean DMax=5.1 cm), 68% exceeded intervention thresholds (>5.5 cm males/>5.0 cm females [25]). Combining DMax>5.5 cm and calcification>50% identified 89% of ruptures in prior studies [11]. However, relying solely on maximum diameter for risk stratification is inadequate, as about 2% of small AAAs rupture annually, while approximately 80% of large aneurysms remain stable [7, 8].

There is a pressing need to identify and validate additional patient-specific diagnostic indicators to better predict which patients are at high risk of aneurysm rupture [29, 30]. Various risk factors, including patient characteristics (e.g., female gender, hypertension, smoking) and aneurysm-related factors (growth rate and rapid increases in intraluminal thrombus), influence AAA development and rupture. However, predicting aneurysm progression and rupture risk remains challenging [31]. Recent studies have explored parameters such as peak wall stress (PWS), peak wall shear stress (PWSS), wall strain, and peak wall rupture index (PWRI) to predict AAA rupture, using methods like finite element analysis (FEA) and fluid-structure interaction (FSI) [32]. While some studies found a significant correlation between these indices and AAA rupture, they often lacked quantitative data [33, 34]. Thus, the clinical applicability and added value of these biomechanical indices compared to maximum diameter remain uncertain. Segmentation of vessels in this study is not without its usefulness. Vessel

segmentation enables automated extraction of DMax, volume, and surface area for biomechanical modeling (e.g., FEA for wall stress [9]). This will be integrated into future rupture risk prediction pipelines.

In recent decades, biomechanical analysis has gained popularity for assessing AAA rupture risk, leading to the development of various biomechanical metrics. However, the results are influenced by factors such as imaging and segmentation methods, wall properties (e.g., tensile strength, intraluminal thrombus, and calcification [16, 35, 36]), and boundary conditions (e.g., pressure [37, 38] and flow [39]). Consequently, further research is needed to establish consistent and clinically valuable metrics.

Previous retrospective studies have shown a positive correlation between calcification and increased AAA rupture risk, supporting the hypothesis that calcification plays a critical role in AAA rupture [11]. Other studies have suggested that abdominal aortic calcification is an independent predictor of cardiovascular events or mortality in the general population [40]. Although calcification scoring and aneurysm diameter are not strongly correlated, combining these factors enhances rupture risk prediction. Specifically, in patients with non-expanding AAA (non-eAAA), a combination of >50% of circumferential vessel calcification and high-risk diameter demonstrates significant predictive value for AAA rupture [11]. In our cohort, the mean maximum AAA diameter was 5.1 ± 1.2 cm, with 68% (68/100) exceeding sex-specific intervention thresholds (≥ 5.5 cm for males; ≥ 5.0 cm for females) [41]. This distribution aligns with contemporary surveillance registries reporting 60–70% of AAAs surpassing intervention size during monitoring. Notably, females exhibited disproportionately higher intervention eligibility (86% vs. males 67%), corroborating established sex-based disparities in AAA behavior where women experience rupture at smaller diameters [26]. These findings reinforce diameter's limitations as a solitary risk predictor - while 32% of our cohort fell below traditional thresholds, 18% of these demonstrated high-risk features (rapid growth >0.5 cm/year or circumferential calcification >50%) [11].

The mean calcification score exceeded 20,000, raising concerns about CTA overestimating density compared to plain CT, influenced by the Agatston scoring system's criteria for calcified lesions [42, 43]. The total Agatston score, designed for coronary artery calcification, may yield higher scores in abdominal aortic aneurysm due to larger plaque volumes. Other vascular calcification assessments include semi-quantitative scores like AAC-8, AAC-24, and the Aortic Calcification Index, which evaluate severity by counting densities in the infrarenal aorta and assigning points based on calcification's relation to vertebral margins [11, 44, 45]. While the Agatston score is widely used for quantifying calcification and predicting

cardiovascular risk, semi-quantitative methods are more complex and less precise, potentially resulting in higher scores in this study [17, 46].

Patient CTA scans segmented the AAA lumen and calcified plaques, as CTA is the gold standard for AAA evaluation, despite non-contrast CT being superior for calcification detection. This technique quantifies AAC non-invasively, aiding surgical planning and reducing radiation exposure [47]. However, extracting AAC regions from CTA is challenging due to contrast medium interference. Future research should correlate CTA-derived Agatston scores with non-contrast CT scores, potentially using phantom calibration for accuracy [48]. The elevated calcification levels in AAA patients suggest a potential link to AAA pathogenesis, which will be a key focus of our research.

Several semi-quantitative computerized methods for segmenting aortic calcification have been developed, some based on the Agatston score [49]. One study compared automated model performance against manual detection, finding that the model successfully identified 96% of calcified regions [50]. However, the clinical applicability of these methods remains uncertain.

We propose a method using nnU-Net for segmenting and detecting abdominal aortic calcification in CTA. Given the strong spatial correlation between the aortic lumen and calcified plaques, we transformed traditional calcification segmentation into a dual-class segmentation problem for both calcification and the aortic lumen. Thrombus (HU:30–80) was excluded by HU threshold (≥ 130). While microcalcifications within thrombus may contribute to rupture risk [51], our method focuses on macroscopic calcification. Histopathological correlation is planned for future work. As demonstrated in femoral tumor segmentation [51], HU thresholding (≥ 130 HU) effectively excludes thrombus despite frequent microcalcifications. Histopathological correlation confirmed such microcalcifications occupy <0.5% plaque volume, explaining their minimal impact on Agatston scores. This aligns with our observation that thrombus-adjacent false positives reduced by 42% with HU constraints. The model achieved a Dice score of 0.924 for aortic segmentation and 0.810 for calcification segmentation. Including lumen labels improved segmentation performance, particularly in low-contrast calcified vessels, likely due to the model's data augmentation capabilities enhancing contrast [52]. Ablation study: Excluding lumen labels reduced calcification Dice by 6.3% (74.7% vs. 81.0%, $p < 0.01$), with significant degradation in low-contrast regions (e.g., posterior aorta). Outliers (Dice <70%) correlated with motion artifacts ($n=3$) or small aneurysms (<3 cm, $n=4$). Bland-Altman analysis revealed slight under-segmentation of calcification (mean area bias: -4.2%). Excluding non-calcified cases may limit generalizability. While our model

achieved Dice > 64% even in low-calcification subregions, future validation should include such cases. Manual verification of calcium presence remains advisable clinically. The $\kappa=0.71$ indicates substantial but imperfect agreement with manual scoring. Discrepancies primarily arose near thresholds (e.g., scores 9.5k–10.5k), where calcium distribution heterogeneity affects categorization. Clinical protocols could mitigate this by recommending repeat CTA for borderline cases (e.g., scores within $\pm 10\%$ of cutoffs). No low risk cases were misclassified as high risk (0% false positives for over-treatment), critical for avoiding unnecessary interventions. However, 10% of high risk cases (4/36) were under-classified (as Moderate), suggesting the model may benefit from incorporating additional features (e.g., plaque morphology) for borderline scores.

Since its development, U-Net has been widely used in medical image-assisted diagnosis and vessel segmentation of a variety of cardiovascular and cerebrovascular diseases. For example, a previous study developed a U-Net-based DL model for automatic segmentation of coronary arteries in invasive coronary angiography with an average DSC of 0.889, demonstrating its efficacy in assisting the diagnosis of coronary artery disease [53]. In a clinical application study of nnU-Net model to assist CTA in the diagnosis of CAS, the nnU-Net model showed good performance in vessel segmentation, with a DSC value of 0.977 in the training set and 0.975 in the validation set, which was also better than the above studies [54]. Similarly, as a newly established algorithm based on the three U-net architectures, the clinical application of the nnU-net model in evaluating the calcification level of abdominal aortic aneurysms using CTA has not been investigated. In this study, the nnU-Net model showed good performance in vessel segmentation, and the Dice value of vessel segmentation in 100 samples reached 0.924, which was numerically better than the above studies [55]. A possible reason why the results of the above studies are inferior to the results of this study is that U-Net has some inevitable shortcomings, including the inability to extract good features, insufficient high-resolution contour information, and asymmetry between edge cut forms and feature images [56]. Secondly, small and medium-sized vessels are highly variable, and are also highly likely to be accompanied by occlusive disease, calcification, intravascular thrombosis, vascular stent graft, or adhesion with surrounding tissues, which may interfere with image analysis. However, nnU-Net further realizes the optimization of data preprocessing, training and data post-processing to avoid the segmentation overlap generated in the segmentation of arterial lumen and calcification contour [57]. Therefore, the vessel segmentation performance of the nnU-Net model is better than that of the U-Net model.

While our five-fold cross-validation design demonstrated robust internal performance metrics, the absence of a fully independent external test set introduces potential for modest scoring bias. Cross-validation may yield optimistic estimates by 3–8% compared to unseen data, particularly in heterogeneous AAA morphologies [58]. To mitigate this, we implemented two safeguards: (1) Strict separation between training/validation folds prevented data leakage, and (2) Our center will prioritize the use of independent test sets and external multicenter datasets to validate the model in the future. Future multicenter validation should confirm generalizability across diverse scanner protocols.

Overall, our nnU-Net model outperformed traditional U-Net models, demonstrating superior vessel segmentation performance and satisfactory consistency with manual segmentation. This enhanced accuracy could facilitate the extraction of other AAA characteristics, such as diameter, length, distance, and volume. Despite challenges posed by imaging artifacts, noise, and incomplete slices, our model achieved good performance in calcified plaque segmentation and maintained consistency with manual evaluation. This suggests that the nnU-Net model is a promising tool for automated AAA calcification assessment, potentially aiding in predicting rupture risk and enhancing clinical decision-making.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12880-025-01911-x>.

Supplementary Material 1

Author contributions

Luo Yuan-lin wrote the main manuscript text and prepared figures. Liu Yi-fan provided technical support, while Huang Zhi, Wang Chu and Zhang Ling-yue assisted in collecting data. Huang Shui-chuan offered ideas for the article and conducted the review. All authors reviewed the manuscript.

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Data availability

To ensure transparency, we clarify that: Patient data cannot be shared publicly due to institutional privacy policies. Non-identifiable data (e.g., model weights, code) are available to facilitate reproducibility. This aligns with BMC's data availability policy.

Declarations

Ethical approval

The studies involving human participants were reviewed and approved by the Medical Ethics Committee of Nanfang Hospital (approval number: NFEC-202504-K3-01). Written informed consent was obtained from all individual participants included in this study. All procedures complied with the Declaration of Helsinki.

Consent for publication

The dataset used in this study was fully anonymized and complied with ethical standards for retrospective analysis. No direct patient identifiers or sensitive clinical details were included in the manuscript.

Data sharing statement

Due to the nature of this research, participants of this study did not agree for their data to be shared publicly, so supporting data is not available.

Competing interests

The authors declare no competing interests.

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